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FIRST SEMESTER B.Tech EXAMINATION, JANUARY-2025

INTRODUCTORY TOPICS IN STATISTICS, PROBABILITY AND CALCULUS

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. **Missing data may be suitably assumed**

ii. **Answer all Questions.**

		Questions	Blooms Level	CO's	Marks
		PART-A	**5 x2=10**		
1.	**a.**	Calculate the mean from the data showing marks of students in a class test: 18, 17, 13, 18, 16.	L3	CO2	**02**
	b.	Find the probability of getting a multiple of 2 or 3 from an unbiased dice.	L3	CO5	**02**
	c.	What is the probability mass function of a Poisson distribution?	L1	CO4	**02**
	d.	Find the derivative of $(3x-2)(2x-3)$ with respect to x.	L2	CO1	**02**
	e.	Evaluate the value of the integral $\int 2x\, sin(x^2)\, dx$.	L2	CO2	**02**
		PART-B	**6 X 5=30**		
2.	**a.**	The mean marks of 100 students were found to be 40. Later on, it was discovered that a score of 53 was misread as 83. Find the correct mean corresponding to the correct score.	L3	CO3	**05**
		OR			
	b.	From the following data compute, the value of the Harmonic Mean:	L3	CO3	**05**

Class interval	10-20	20-30	30-40	40-50	50-60
Frequency	4	6	10	7	3

3.	**a.**	The probability that a regularly scheduled flight departs on time is $P(D) = 0.83$; the probability that it arrives on time is $P(A) = 0.82$; and the probability that it departs and arrives on time is $P(D \cap A) = 0.78$. Find the probability that a plane i. arrives on time, given that it departed on time, ii. departed on time, given that it has arrived on time.	L3	CO5	**05**
		OR			
	b.	In a class, 70% are boys and 30% are girls. 5% of boys and 3% of girls are irregular in their classes. What is the probability that a student selected at random is irregular to the classes and what is the probability that the irregular student is a girl?	L3	CO4	**05**
4.	**a.**	If X is Uniform between 8 and 20 with a mean of 14 and variance of 12, then find $P(\lvert X-6\rvert \le 15)$ and $P(15 \le X \le 17)$.	L3	CO6	**05**
		OR			

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	b.	Suppose that a large conference room at a certain company can be reserved for no more than 4 hours. Both long and short conferences occur quite often. Assume that the length X of a conference has a uniform distribution on the interval [0, 4]. i. What is the probability density function? ii. What is the probability that any given conference lasts at least 3 hours?	L3	CO6	**05**
5.	**a.**	If $x = 2at, y = at^2$, then show that $\frac{dy}{dx} - \frac{2y}{x} = 0$	L-3	CO1	**05**
		OR			
	b.	If $x^y = e^{x-y}$, then show that $\frac{dy}{dx} = \frac{\log x}{(1+\log x)^2}$.	L-3	CO1	**05**
6.	**a.**	Evaluate $\int \frac{2(x+3)}{(x+2)(x+4)}\, dx$	L3	CO2	**05**
		OR			
	b.	Evaluate $\int\int xy\, dxdy$ over the positive quadrant of the circle $x^2 + y^2 = a^2$.	L3	CO2	**05**
7.	**a.**	Explain the different methods of graphical representation of statistical data.	L2	CO2	**05**
		OR			
	b.	Find the value of $\frac{dy}{dx}$ for the implicit function $2x^2y + 3x^3y^2 + 4y^2x^3 + 2xy = 14$	L3	CO1	**05**
		PART-C	**6 X 10=60**		
8.	**a.**	Determine the mode weight from the following data of 122 persons given as follows.	L3	CO4	**10**

Weight (in lbs)	100-110	110-120	120-130	130-140	140-150	150-160	160-170	170-180
No.of persons	4	6	20	32	33	17	8	2

		OR						
	b.	Calculate the lower and upper quartiles, from the following data:	L3	CO4	**10**

Central Value	2.5	7.5	12.5	17.5	22.5
Frequency	7	18	25	30	20

9.	**a.**	Four units of a bulb manufacturing factory produce defective bulbs at rates of 3%, 2%, 1%, and 0.5%, respectively. A randomly selected bulb from the factory's total output is found to be defective. What is the probability that it was produced by the fourth unit of the factory?	L3	CO5	**10**
		OR			
	b.	Three urns I, II, and III contain respectively 2 white and 3 black balls, 3 white and 4 black balls, and 4 white and 5 black balls. One of the urns is chosen randomly, the probability of each urn chosen being equal. Then one ball is drawn at random from the urn chosen and found to be black. What is the probability that urn I, II, or III was chosen?	L3	CO5	**10**
10.	**a.**	Ten coins are tossed simultaneously. Find the probability of getting (i) at least seven heads and (ii) exactly seven heads.	L3	CO4	**10**
		OR			
	b.	A random variable X is normally distributed with mean 9 and standard deviation 3. Find $P(X \leq 15), P(X \geq 15)$ and $P(0 \leq X \leq 9)$. Given $\phi(Z \leq 2) = 0.9772$, $\phi(Z \leq 0) = 0.5$, $\phi(Z \leq -3) = 0.0013$.	L3	CO5	**10**
11.	**a.**	i. If $y = e^x \log(x)$ then show that $xy'' - (2x-1)y' + (x-1)y = 0$ ii Differentiate $\sin^2 x$ with respect to x^2.	L3	CO1	**10**
		OR			
	b.	Find the maximum and minimum value of the function $f(x) = 3x^3 - 9x^2 - 27x + 30$.	L3	CO1	**10**
12.	**a.**	Find the value of the integral $\int_0^1 \int_x^{\sqrt{x}} (x^2 + y^2)\,dy\,dx$.	L3	CO2	**10**
		OR			
	b.	Evaluate the integral $\int x^3 e^x dx$	L3	CO2	**10**
13.	**a.**	Find the value of the integral $\int_{-1}^{1} \int_0^z \int_{x-z}^{x+z} (x+y+z)\,dy\,dx\,dz$	L3	CO2	**10**
		OR			
	b.	Let X be a discrete random variable with the following probability distribution	L3	CO6	**10**

x	0	1	2	3
P(x)	0.15	0.38	0.10	0.01

Find (i) Mean , (ii) Variance and (iii) Standard Deviation